YOUSEF KHALED GOUDA

Senior AI Systems Architect · Principal ML Engineer · AI Research Lead

20+ Years Building Production AI Systems · LLMs · MLOps · Computer Vision · NLP · Arabic AI

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21 Years	200+	Arabic / EN	15+ Papers	Architect
Industry Experience	Production AI Projects	Bilingual AI Expert	Research & Publications	Senior AI Systems Design

EXECUTIVE SUMMARY

Senior AI Systems Architect and Principal Machine Learning Engineer with over 21 years of hands-on experience designing, building, and scaling production AI infrastructure across Egypt, the Gulf, and international remote engagements. Deep expertise spanning the full AI lifecycle — from classical machine learning and statistical modeling (2005–2012), through the deep learning revolution (2012–2018), to modern large language model systems, RAG architectures, and autonomous AI agents (2018–present). Recognized authority on Arabic NLP and multilingual AI systems; author of 15+ technical research reports; architect of AI platforms handling millions of daily inferences. Currently focused on LLM fine-tuning, AI agent orchestration, interpretability research, and MLOps at enterprise scale.

EDUCATION

B.Sc. — Computer & Control Systems Engineering

Sep 1999 – Jun 2004

Faculty of Engineering · Benha University · Banha, Egypt · Graduated with Honours

- Thesis: 'Adaptive Control Systems for Industrial Process Automation Using Fuzzy Logic and Neural Networks' — Grade: Excellent.
- Core studies: Control Theory, Digital Signal Processing, Electronics, Algorithms, Computer Architecture, and Embedded Systems.
- Final-year project: PID auto-tuning system using a neural network controller — implemented in C and tested on AVR hardware.

PROFESSIONAL EXPERIENCE

2022 – Present | LLM Era · Foundation Models · Autonomous Agents

Senior AI Systems Architect — Principal

Jan 2022 – Present

Freelance / Remote · International Clients (UAE, Saudi Arabia, Europe) · Cairo Base

- Architect of enterprise LLM platforms: designed and deployed RAG systems, fine-tuned instruction-following Arabic LLMs (LLaMA 2/3, Mistral), and built multi-agent orchestration frameworks for 8+ clients.
- Led Arabic LLM alignment project: curated 80K Arabic instruction dataset, applied QLoRA + DPO pipeline, improved human preference accuracy from 54% to 82% — deployed to production via vLLM on GCP.
- Designed a retrieval-augmented legal document QA platform for an Egyptian law firm: LangChain + Qdrant + cross-encoder reranking; serving 3,000+ daily queries at P95 latency < 900ms.
- Established MLOps standards across client organizations: Docker, GitHub Actions CI/CD, MLflow experiment registry, Weights & Biases dashboards, and Grafana/Prometheus production monitoring.
- Delivered 15+ technical workshops on LLM architecture, prompt engineering, and AI system design for engineering teams in Egypt and Gulf countries.
- Published 6 technical research reports (2022–2026) on Arabic LLM evaluation, hallucination detection, RLHF data quality, and RAG system benchmarking.

2018 – 2022 | Transformer Era · BERT · GPT · NLP Scale-Up

Lead AI Engineer & Research Manager

Mar 2018 – Dec 2021

Egyptian Company for Software & AI Solutions (ECSA) · Cairo, Egypt

- Led a team of 9 ML engineers building NLP and computer vision products for Arabic-language markets across Egypt, Saudi Arabia, and the UAE.

- Architected and productionized an Arabic text classification platform (10M+ documents/day) using fine-tuned BERT multilingual, later migrated to AraBERT — accuracy improved from 81% to 93%.
- Built an Arabic speech recognition pipeline: integrated Wav2Vec 2.0 fine-tuned on Egyptian dialect corpus (200 hours); deployed as a REST API serving 50K+ daily transcriptions.
- Designed a real-time object detection and tracking system for a smart-city surveillance contract: YOLOv4 + DeepSORT; processing 30 camera streams simultaneously on 4× V100 GPU cluster.
- Introduced Transformer-based models to the team in 2018 when BERT was first released — led internal study group and migrated 3 production NLP pipelines to BERT within 6 months.
- Established the company's first MLOps infrastructure: Kubernetes-based model serving, A/B testing framework, automated retraining pipelines, and model drift monitoring.

2014 – 2018 | Deep Learning Revolution · CNNs · RNNs · Keras · TensorFlow

Senior Machine Learning Engineer

Jun 2014 – Feb 2018

Softex Egypt / Information Technology Industry Development Agency (ITIDA) Projects · Cairo

- Pioneered adoption of deep learning in Egypt's public-sector AI initiatives — among the first engineers in Egypt to deploy CNNs to production (2015) for document digitization and automated form recognition.
- Built an Arabic handwriting recognition system (CNN + CTC loss) for Egypt's national ID digitization project — processing 2M+ ID card images; character accuracy 96.2%.
- Developed license plate recognition for Egyptian traffic authority: custom CNN trained on 50K Egyptian plates in Theano/Keras; 94% recognition accuracy in real traffic conditions.
- Led migration from Theano to TensorFlow (2015–2016) across 5 active projects as the ecosystem shifted; authored internal migration guide used by 30+ engineers.
- Built RNN/LSTM-based Arabic text generation and autocomplete system — early Arabic NLP product deployed in a government document management platform.
- Mentored 6 junior engineers; introduced weekly deep learning paper reading sessions that became a company-wide practice.

2010 – 2014 | Classical ML · Statistical Modeling · Big Data Emergence

Machine Learning Engineer — Mid–Senior Level

Jan 2010 – May 2014

Raya Contact Center / Raya IT · Cairo & Alexandria, Egypt

- Designed and deployed ML-powered predictive analytics systems for Egypt's largest contact center operator: churn prediction (XGBoost, 88% AUC), call volume forecasting (ARIMA + gradient boosting), and agent performance scoring.
- Built a real-time fraud detection pipeline using anomaly detection (Isolation Forest, One-Class SVM) on call records — reduced fraudulent call losses by 34% in first year.
- Led data science team of 4 engineers; introduced reproducible ML workflows with versioned datasets, model registries, and automated evaluation reports.
- Worked extensively with Python (2.7/3.x), R, Pandas, NumPy, Scikit-learn, and SQL on datasets ranging from 10M to 500M rows.
- Delivered Arabic sentiment analysis v1 using SVM + TF-IDF on customer feedback transcripts — company's first NLP product (2012).

2005 – 2010 | Foundations · Software Engineering · Early Data Science

Software Engineer → Data Analyst

Jul 2005 – Dec 2009

Xceed Contact Center Egypt / Link Development · Cairo, Egypt

- Began career as a software engineer building C++ and Python data processing tools for large-scale call-center analytics platforms.
- Developed statistical reporting systems and early predictive models using logistic regression and decision trees in R and Python.
- Implemented automated data pipelines processing 5M+ daily records; reduced manual reporting effort by 70%.
- Self-taught machine learning from foundational textbooks (Bishop PRML, Hastie ESL) while applying techniques to real business problems.
- Earned Oracle Database and Microsoft certifications; led SQL optimization efforts reducing query execution time by 60% across reporting systems.

CERTIFICATIONS & CONTINUOUS LEARNING

2024 – 2026 | Agents · Advanced LLM Systems · AI Safety

- AI Agents in LangGraph — DeepLearning.AI + LangChain (2024)
- Multi AI Agent Systems with crewAI — DeepLearning.AI (2024)
- Building Agentic RAG with LlamaIndex — DeepLearning.AI (2024)
- Mechanistic Interpretability — ARENA (Neel Nanda) (2024)
- AI Safety Fundamentals: Alignment Course — BlueDot Impact (2024)
- Efficiently Serving LLMs — DeepLearning.AI (2024)
- Multimodal RAG: Chat with Videos — DeepLearning.AI (2025)
- JavaScript AI SDK & Vercel AI — Vercel / deeplearning.ai (2025)

2022 – 2023 | LLMs · RLHF · MLOps · RAG Systems

- LangChain for LLM Application Development — DeepLearning.AI (2023)
- Building Systems with the ChatGPT API — DeepLearning.AI (2023)
- Finetuning Large Language Models — DeepLearning.AI (2023)
- Building & Evaluating Advanced RAG — DeepLearning.AI + LlamaIndex (2023)
- RL from Human Feedback (RLHF) — DeepLearning.AI (2023)
- MLOps Specialization (4 courses) — DeepLearning.AI / Coursera (2022)
- Full Stack Deep Learning — FSDL (fullstackdeeplearning.com) (2022)
- Hugging Face NLP Course — Hugging Face (huggingface.co/learn) (2022)
- Hugging Face Diffusion Models Course — Hugging Face (2022)
- Practical Deep Learning for Coders Part 2 — fast.ai (2022)

2019 – 2021 | Transformers · BERT · NLP at Scale

- Natural Language Processing Specialization (4 courses) — DeepLearning.AI / Coursera (2019)
- CS224N: NLP with Deep Learning — Stanford Online (Audit) (2019)
- TensorFlow Developer Professional Certificate — Google / Coursera (2020)
- Applied AI with DeepLearning — IBM / Coursera (2020)
- Practical Deep Learning for Coders (Part 1) — fast.ai / Jeremy Howard (2019)
- Transformer Models & BERT Model — Google Cloud (Coursera) (2020)
- Computer Vision Specialization — DeepLearning.AI / Coursera (2021)
- ML Pipelines on Google Cloud — Google Cloud Training (2021)

2016 – 2018 | Deep Learning Maturity · PyTorch · Sequence Models

- Deep Learning Specialization (5 courses) — Andrew Ng / DeepLearning.AI (2017)
- CS231n: CNNs for Visual Recognition — Stanford Online (Audit) (2016)
- CS285: Deep Reinforcement Learning — UC Berkeley (Audit) (2018)
- PyTorch Scholarship Challenge — Facebook AI / Udacity (2018)
- Advanced Machine Learning Specialization — HSE Moscow / Coursera (2018)
- Sequence Models — Andrew Ng / DeepLearning.AI / Coursera (2018)

2012 – 2015 | Foundations Era · Early MOOCs · Scikit-learn · Theano

- Machine Learning — Andrew Ng / Stanford / Coursera (2012, original course)
- Statistical Learning — Stanford Online (Trevor Hastie & Robert Tibshirani) (2013)
- Neural Networks for Machine Learning — Geoffrey Hinton / Coursera (2013)
- Data Science Specialization (10 courses) — Johns Hopkins / Coursera (2014)
- Mining Massive Datasets — Stanford / Coursera (2013)
- Probabilistic Graphical Models — Daphne Koller / Stanford / Coursera (2013)
- Algorithms: Design & Analysis — Tim Roughgarden / Stanford / Coursera (2012)
- Computing for Data Analysis (R) — Roger Peng / Johns Hopkins / Coursera (2013)

2005 – 2011 | Professional Foundations · CS & Engineering Certifications

- Oracle Database 10g: SQL Fundamentals & PL/SQL — Oracle Certified Associate (2006)
- Microsoft Certified: MCP — Programming in C# .NET (2007)
- CCNA: Cisco Certified Network Associate — Cisco (2008)
- R Programming — Self-study / Coursera early access (2011)
- Linux Professional Institute Certification (LPIC-1) — LPI (2007)
- Project Management Fundamentals — PMI / Cairo Chapter (2009)
- Pattern Recognition & Machine Learning — Bishop (Self-study, 2009–2011)
- Elements of Statistical Learning — Hastie, Tibshirani (Self-study, 2010)

RESEARCH & TECHNICAL PUBLICATIONS

All technical reports published on GitHub (github.com/yousefkhaled2005) and Hugging Face Hub. Selected reports presented at Egyptian and regional AI meetups.

2023 – 2026 | LLM Research

Arabic LLM Evaluation Framework — Hallucination & Factuality Benchmark *2025*

GPT-4o · LLaMA 3 · Mistral · Gemini · Claude · 1,200-Query Arabic Test Suite

- Comprehensive evaluation of 6 frontier models on Arabic factual QA (legal, medical, historical domains); custom 3-tier hallucination taxonomy.
- Showed 28% higher hallucination rate in Arabic vs. English for same models; results submitted to Arabic NLP Workshop proceedings.

RLHF Preference Data Quality — Impact on Arabic LLM Alignment *2024*

DPO · TRL · 1,000-Pair Annotated Dataset · Kappa Agreement Analysis

- Empirical study on how annotation agreement thresholds affect downstream DPO alignment; showed Kappa ≥ 0.78 is required for consistent alignment gains.
- Practical annotation guidelines for Arabic RLHF projects; adopted by 2 Gulf-based AI teams as their internal standard.

RAG vs. Fine-Tuning for Arabic Domain QA — Comparative Benchmark *2023*

LangChain · Qdrant · QLoRA · Arabic Legal & Medical Corpora · 500-Query Eval

- Systematic comparison across 4 domains; RAG superior for factual recall (88% vs 84%), fine-tuning superior for latency (3× faster); hybrid architecture proposed.

Efficient Arabic LLM Serving — Inference Engine Benchmark *2023*

vLLM · llama.cpp · GGUF INT4/INT8 · Triton · T4/A100 GPU · CPU Benchmarks

- Head-to-head benchmark of vLLM, llama.cpp, and Triton on LLaMA 2/3 variants; comprehensive throughput, latency, and memory analysis across 6 hardware configs.

2018 – 2022 | Deep Learning & NLP Research

AraBERT vs. Multilingual BERT — Systematic Benchmark for Arabic NLP *2020*

Hugging Face · AraBERT · CAMeLBERT · 6 Benchmarks · 4 Task Types

- First systematic Arabic BERT comparison covering sentiment, NER, QA, and dialect ID; AraBERT-v2 outperforms mBERT by 4–11 F1 points across tasks.
- Widely cited in Egyptian AI community; report used as baseline by 3 university NLP research groups.

Arabic Dialect Identification at Scale — Dataset & Baseline Study *2019*

CAMeL Tools · MADAR · 5-Class Classification · Cross-dialect Evaluation

- Created and released a 5,000-sentence Arabic dialect evaluation dataset (Egyptian, Gulf, Levantine, Maghrebi, MSA) with careful annotation guidelines.
- Baseline models and annotation guide published on GitHub; dataset downloaded 800+ times on Hugging Face Hub.

YOLOv4 vs. EfficientDet for Arabic OCR Scene Text Detection *2020*

TensorFlow · YOLOv4 · Custom Arabic Scene Text Dataset · mAP Evaluation

- Compared detection architectures on Arabic scene text (storefronts, road signs) — YOLOv4 achieved mAP@0.5 = 0.79 vs. EfficientDet-D3 at 0.77 with 2× faster inference.

2013 – 2017 | Classical ML & Early Deep Learning Research

Ensemble Methods for Arabic Sentiment Analysis — Pre-Transformer Era *2015*

Python · Scikit-learn · SVM · Random Forest · TF-IDF · Arabic Text Corpus

- Systematic comparison of 8 classical ML models on Arabic sentiment; SVM + TF-IDF bigrams achieved 86.4% accuracy — published as internal technical report at Softex Egypt.

Deep Neural Networks for Egyptian Arabic ASR — Early Study *2017*

Theano · Keras · LSTM · CTC Loss · Custom 50-Hour Egyptian Dialect Corpus

- First deep learning ASR experiment on Egyptian Arabic dialect at the company; LSTM+CTC achieved 34% WER — 18% improvement over GMM-HMM baseline.

SELECTED PRODUCTION AI PROJECTS (200+ TOTAL)

2022 – 2026 | LLM & Agent Systems

Enterprise Arabic LLM Platform — Fine-Tuning + Deployment *2023–2024*

LLaMA 3 8B/70B · QLoRA · Axolotl · DPO · TRL · vLLM · GCP · Qdrant · LangChain

- End-to-end Arabic LLM: curated 80K instruction samples, QLoRA fine-tuning on 8×A100, DPO alignment with 1K preference pairs; deployed via vLLM on GCP with auto-scaling.
- Production system serving 50K+ daily Arabic queries; P95 latency 420ms; 82% human preference score in blind evaluation vs. GPT-3.5-turbo at 71%.
- Full MLOps stack: MLflow registry, W&B training dashboard, Langfuse observability, Prometheus/Grafana production monitoring.

Multi-Agent AI Research & Report Generation System *2024*

LangGraph · Claude API · ArXiv API · SerpAPI · FastAPI · React · Mem0

- Autonomous research agent: searches ArXiv + web, parses papers, synthesizes cross-source insights, and generates structured technical reports in Arabic and English.
- Stateful LangGraph orchestration with tool use, episodic memory, and human-in-the-loop approval for high-stakes actions; deployed as a full-stack web app.

Production RAG — Arabic Legal Document Intelligence *2023*

LangChain · Qdrant · Multilingual-E5 · Cross-encoder Reranking · FastAPI · Docker

- RAG platform over 10K+ Arabic legal documents; hybrid BM25 + dense retrieval + cross-encoder reranking; 88% accuracy on 500-query gold standard.
- Containerized microservice architecture; GCP Cloud Run deployment; 3,000+ daily queries at P95 < 900ms latency.

2018 – 2022 | Deep Learning & Computer Vision Systems

Arabic Text Classification Platform — 10M Documents/Day *2018–2019*

AraBERT · TensorFlow Serving · Kubernetes · Redis · PostgreSQL · REST API

- Production NLP platform classifying Arabic news, social media, and government documents into 12 categories at 10M+ documents/day throughput.
- Migrated from SVM+TF-IDF (81% accuracy) to AraBERT (93%); Kubernetes autoscaling; Redis response caching; zero-downtime rolling deployments.

Smart-City Surveillance — Real-Time Multi-Camera Tracking *2019–2020*

YOLOv4 · DeepSORT · OpenCV · TensorRT · NVIDIA V100 Cluster · RTSP Streams

- Multi-object detection and tracking across 30 simultaneous RTSP camera streams; YOLOv4 + TensorRT INT8 at 28 FPS per stream on 4×V100 cluster.
- Real-time alert system for crowd density, loitering, and perimeter breach; integrated with city operations dashboard via WebSocket streaming.

Arabic Speech Recognition API — Egyptian Dialect *2020–2021*

Wav2Vec 2.0 · Hugging Face · 200h Egyptian Corpus · FastAPI · Redis · Docker

- Fine-tuned Wav2Vec 2.0 on 200-hour Egyptian Arabic speech corpus (call center recordings); 19% WER — best-in-class for Egyptian dialect at the time.
- REST API serving 50K+ daily transcriptions; async processing queue with Redis; auto-scaling Docker deployment on AWS ECS.

2012 – 2018 | Classical ML & Early Deep Learning

National ID Digitization — Arabic OCR Pipeline *2015–2016*

Keras (Theano backend) · CNN · CTC Loss · 2M+ ID Images · Python · PostgreSQL

- CNN-based Arabic handwriting OCR for Egypt's national ID digitization program; 96.2% character accuracy on 2M+ scanned ID cards.

- One of Egypt's first production CNN deployments; custom data augmentation pipeline for handling handwriting variability and scan quality issues.

Contact Center Fraud Detection & Churn Prediction System

2010–2013

Python · Scikit-learn · XGBoost · ARIMA · SQL · Hadoop (early cluster)

- Real-time fraud detection using Isolation Forest + One-Class SVM on call records; reduced fraudulent call losses by 34% in year one.
- Churn prediction model (XGBoost, AUC = 0.88) deployed to guide retention campaigns; directly attributed to 12% improvement in customer retention rate.
- Call volume forecasting (ARIMA + gradient boosting) improved staffing efficiency by 23% — one of Egypt's first ML-driven contact center optimization systems.

OPEN SOURCE & GITHUB PORTFOLIO

github.com/yousefkhaled2005 | huggingface.co/yousef-khaled — 20+ public repositories · 8 Hugging Face models · 3 public datasets

◆ Models Published on Hugging Face Hub

- yousef-khaled/arabert-sentiment-v2 — Arabic/English sentiment (F1=0.914)
- yousef-khaled/arabic-dialect-classifier — 5-class dialect ID (F1=0.87)
- yousef-khaled/llama3-arabic-instruct — QLoRA fine-tuned Arabic LLM
- yousef-khaled/arabic-hallucination-detector — LLM output factuality
- yousef-khaled/arabic-ner-v2 — NER model for Arabic (PER/ORG/LOC/DATE)
- yousef-khaled/wav2vec2-egyptian-arabic — ASR for Egyptian dialect
- yousef-khaled/arabic-dialect-eval-1k — Egyptian Arabic eval benchmark
- yousef-khaled/arabic-rlhf-preferences — 1K-pair preference dataset

◆ Key GitHub Repositories

- arabic-llm-platform — Full QLoRA+DPO fine-tuning pipeline (Axolotl)
- production-rag-arabic — LangChain+Qdrant RAG for Arabic legal docs
- multi-agent-research — LangGraph autonomous research agent
- arabic-asr-pipeline — Wav2Vec 2.0 Egyptian dialect ASR system
- llm-inference-benchmark — vLLM vs llama.cpp vs Triton comparison
- arabic-bert-benchmark — AraBERT vs mBERT systematic study
- arabic-dialect-dataset — MADAR fine-tuning + 1K Egyptian eval set
- yolov4-arabic-ocr — Scene text detection for Arabic storefronts
- rlhf-annotation-tools — Python toolkit for preference data QA
- ml-system-design — Architecture patterns for production AI systems

TECHNICAL SKILLS

LLMs & Generative AI (Expert)

LLaMA 2/3 · Mistral · Mixtral MoE · GPT-4o · Claude 3.5 · Gemini — Fine-tuning (LoRA, QLoRA, Full SFT) · RLHF / DPO / GRPO · Prompt Engineering · RAG · Function Calling · LangChain · LlamaIndex · LangGraph · DSPy · Hugging Face PEFT & TRL · Axolotl

Machine Learning & Deep Learning (Expert)

PyTorch · TensorFlow · Keras · JAX/Flax · Scikit-learn · XGBoost · LightGBM · CNNs · RNNs / LSTMs · Transformers · Attention Mechanisms · Diffusion Models · VAEs · GANs — 21 years applied experience from classical ML through modern foundation models

Computer Vision (Expert)

YOLOv4/v5/v8 · SAM (Segment Anything) · OpenCV · TensorRT · ONNX Runtime · ResNet / EfficientNet / ViT · CLIP · Object Detection / Segmentation / Tracking (DeepSORT) · OCR (Arabic & English)

NLP & Arabic AI (Expert)

AraBERT · CAMeLBERT · Multilingual-E5 · CAMeL Tools · Arabic dialect processing · NER / POS / Sentiment / QA · Bilingual pipeline design · ASR (Wav2Vec 2.0 Egyptian Arabic) · TTS

MLOps & Infrastructure (Expert)

Docker · Kubernetes · GCP (Cloud Run, Vertex AI) · AWS (ECS, SageMaker) · GitHub Actions CI/CD · MLflow · Weights & Biases · Langfuse · Prometheus + Grafana · vLLM · Triton Inference Server · TensorRT · Redis · FastAPI · Celery

Vector Search & Databases

Qdrant · Weaviate · FAISS · pgvector · Pinecone · BM25 · Cross-encoder Reranking · PostgreSQL · MongoDB · Elasticsearch

Programming

Python (Expert, 20+ years) · C++ (Advanced, 18+ years) · C (Advanced) · SQL (Expert) · Bash / Linux CLI · R · JavaScript / TypeScript · CUDA (intermediate)

LANGUAGES

Arabic
Native — Fluent (Speaking, Reading, Writing)
Egyptian dialect + MSA · 21 years Arabic AI/NLP work

English
Professional Working Proficiency (B2–C1)
All technical documentation, research & client communication